

Unifying Special Relativity and Quantum Mechanics

Lecture 31 — The Electroweak Spectrum:

$W^\pm$, Z , the Photon, and the Bosonic Interactions

(diagonalising the W^3 – B mass matrix; the Weinberg angle and the mass eigenstates $Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu$, $A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu$; why one eigenvalue is exactly zero; electric charges from the unbroken $U(1)$; the mode expansion of $W^\pm$; the mass ladder M_W , M_Z , M_h ; the three categories of bosonic interaction; then: the opening of the fermionic sector and chirality)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–30: normal text and the yellow / pink / grey / green boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil, brown). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination — for once, this one is simple.

1. **No duplicated sheets.** This is the first scan in a long while that does not repeat the tail of the previous one. Lecture 30 ended at [p. 6*****]; this scan opens at [p. 7*****] and runs straight through to [p. 18*****]. PDF page n is sheet $n + 6$, with no gaps, no blanks, and no chapter restarts.
2. **Same chapter throughout,** so every margin marker carries the same *fifteen* stars: [p. 7*****]–[p. 18*****]. The chapter is still *A Standard Model primer*, begun on [p. 1*****] in Lecture 30.

It stops mid-development, as usual — and this time in the most tantalising place yet. The last sheet, [p. 18*****], finishes the bosonic sector, opens a new section headed *The fermionic sector*, defines the chiral projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$, and breaks off at the sentence “the left and right components transform differently under $SU(3)_c \times SU(2)_L \times U(1)_Y$ ”. Section 13 carries that forward: the representation table, why fermion mass terms are *forbidden*, the Yukawa couplings, and anomaly cancellation.

A note on notation — the $SU(2)_L$ coupling changes name here. Lecture 30 wrote the weak coupling as g ; from [p. 7*****] onwards the notes write g_L , reserving g_Y for hypercharge as before. The two are the same object; g_L is used throughout this document to match the sheets. So $M_W = \frac{1}{2}g_L v$ here is the same formula as $m_W = \frac{1}{2}g v$ there.

Slips and reading notes, flagged in place.

1. On [p. 9*****] the two cancelling $v^4 g_L^2 g_Y^2 / 16$ terms are struck out in the original with a diagonal line; they cancel exactly, which is precisely why one eigenvalue vanishes. Kept as an explicit cancellation here.
2. On [p. 9*****] the diagonal matrix is written $M_\Delta^2 = \text{diag}(\frac{v^2}{4}(g_L^2 + g_Y^2), 0)$ — i.e. with the *massive* eigenvalue first. That ordering fixes the row order of O on [p. 10*****] and hence which combination is called Z and which is called A . Worth keeping straight.
3. On [p. 10*****] the eigenvector components are written (V, W) — these are just labels for the two entries, not fields. Renamed (a, b) here to avoid collision with W_μ^3 .
4. On [p. 16*****] the identity $[(W_\mu^1)^2 + (W_\mu^2)^2] = 2W_\mu^+ W_\mu^{-,\mu}$ is used without proof; verified in a blue box there.

5. On [p. 14*****] the mode expansion carries three labels on the operators, $a(\vec{p}, \lambda, +1)$: momentum, helicity, and *charge*. The third is the new one, and it is the whole point of the sheet.

Notation. g_L the $SU(2)_L$ coupling, g_Y the $U(1)_Y$ coupling, θ_W the Weinberg (weak mixing) angle, $v^2 = \mu^2/\lambda$. W_μ^A ($A = 1, 2, 3$) the $SU(2)_L$ gauge fields, B_μ the hypercharge field; $W_\mu^\pm$, Z_μ , A_μ the mass eigenstates. $T^A = \sigma^A/2$ acting on the doublet, t^A the adjoint generators ($(t^A)_{BC} = -i\varepsilon_{ABC}$), Y hypercharge, $Q = T^3 + Y$.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 30, and a note on the added section there

Lecture 30 built the whole non-abelian Higgs mechanism and then applied it to the Standard Model. The chain, compressed:

- **The general theorem.** Evaluating $\mathcal{L}$ on the vacuum isolates $M_{\alpha\beta}^2 = 2g^2\Phi_0^\dagger X^\alpha X^\beta \Phi_0$, a *Gram matrix* of the vectors $X^\alpha \Phi_0$. It is hermitian and positive definite, so exactly $\dim G - \dim H$ gauge bosons become massive and $\dim H$ stay exactly massless.
- **Two worked examples.** $SU(2) \rightarrow 1$ with a doublet (all three generators break; three degenerate W 's; the Higgs unitarises $W_L W_L$ scattering), and the Georgi–Glashow model $SU(2) \rightarrow U(1)$ with an adjoint triplet (one $U(1)$ survives; $W^\pm$ of charge ± 1 and a massless W^3).
- **The Standard Model.** $SU(3)_c \times SU(2)_L \times U(1)_Y$ with $H \sim (1, 2, \frac{1}{2})$. No single generator annihilates H_0 , but $T^3 H_0$ and $Y H_0$ are equal and opposite, so $Q = T^3 + Y$ survives: $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$, three Goldstones. Expanding $(D_\mu H)^\dagger (D^\mu H)$ produced the quadratic gauge terms including a $W_\mu^3 B^\mu$ *cross term* — and the scan stopped there.

A check worth recording. Section 18 of the Lecture 30 document carried that diagonalisation through as *added* material, obtaining

$$M_W = \frac{1}{2}gv, \quad A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu \quad (m = 0),$$

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad M_Z = \frac{v}{2} \sqrt{g^2 + g_Y^2},$$

with $\tan \theta_W = g_Y/g$ and $\rho = 1$. Sheets [p. 7*****]–[p. 11*****] of *this* scan do exactly that calculation, and the results agree line for line (with $g \rightarrow g_L$). The added completion was correct; what follows is the author's own version of it, in more detail.

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PART I — Diagonalising the neutral sector [p. 7***]–[p. 11*****]**

1 Separating $W^{1,2}$ from W^3

[p. 7*****]

It is convenient to separate $W_\mu^{1,2}$ from W_μ^3 , since the latter **mixes** with B_μ

[p. 7*****]

$$\begin{aligned} & \frac{g_L^2}{8} (h+v)^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu}) + \frac{g_L^2}{8} (h+v)^2 W_\mu^3 W^{3,\mu} \\ & + \frac{g_Y^2}{8} (v+h)^2 B_\mu B^\mu - \frac{g_L g_Y}{4} (h+v)^2 W_\mu^3 B^\mu = \\ & = \frac{g_L^2}{8} (h+v)^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu}) + \frac{(h+v)^2}{8} (g_Y B_\mu - g_L W_\mu^3)^2 \end{aligned}$$

We find

$$\begin{aligned} (D_\mu H)^\dagger (D^\mu H) &= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \underbrace{\frac{g_L^2}{8} (h+v)^2 [(W_\mu^1)^2 + (W_\mu^2)^2]}_{(2)} \\ &+ \underbrace{\frac{1}{8} (v+h)^2 (g_Y B_\mu - g_L W_\mu^3)^2}_{(1)} \end{aligned}$$

(1) We extract

(2) We extract

$$\frac{v^2}{8} (g_Y B_\mu - g_L W_\mu^3)^2$$

$$\frac{1}{2} \left(\frac{g_L^2 v^2}{4} \right) [(W_\mu^1)^2 + (W_\mu^2)^2]$$

⇓

⇓

We cannot read immediately the masses

W^1 and W^2 have mass

$$M_W^2 = \frac{1}{4} g_L^2 v^2$$

Added explanation (not in the original notes) — Completing the square — the algebraic step that makes the whole page work

The three neutral terms are collected into a perfect square. Written out:

$$\frac{g_L^2}{8} W^3 W^3 - \frac{g_L g_Y}{4} W^3 B + \frac{g_Y^2}{8} B B = \frac{1}{8} \left[g_L^2 W^3 W^3 - 2 g_L g_Y W^3 B + g_Y^2 B B \right] = \frac{1}{8} (g_Y B_\mu - g_L W_\mu^3)^2.$$

The cross term's coefficient is exactly $-\frac{1}{4} g_L g_Y$ — i.e. exactly $-2 \times$ the geometric mean of the two diagonal coefficients $\frac{1}{8} g_L^2$ and $\frac{1}{8} g_Y^2$. That is the algebraic condition for a quadratic form to be a *perfect square*, and a perfect square in two variables is a rank-one form: it has one zero eigenvalue.

So the massless photon is already visible here, before any matrix is written down. The combination $g_Y B_\mu - g_L W_\mu^3$ appears; the orthogonal combination $g_L B_\mu + g_Y W_\mu^3$ does not appear *at all*, and therefore has no mass term. Sheets [p. 8*****]–[p. 10*****] are the careful, systematic version of this one-line observation.

Why $W^{1,2}$ can be read off immediately but W^3 cannot. W^1 and W^2 appear only in the combination $(W^1)^2 + (W^2)^2$ — diagonal, with equal coefficients. W^3 appears mixed with B . The comparison with the Proca normalisation $\frac{1}{2} M^2 W_\mu W^\mu$ then gives $M_W^2 = \frac{1}{4} g_L^2 v^2$ directly, matching Example 1 of Lecture 30 [p. 19*****] exactly, as it must — the charged sector never sees hypercharge.

2 The mass matrix in matrix form

[p. 8*****]

Let us write in matrix form the term $\frac{v^2}{8} (g_Y B_\mu - g_L W_\mu^3) (g_Y B^\mu - g_L W^{3,\mu})$. We have

[p. 8*****]

$$\frac{1}{2} (W_\mu^3, B_\mu) \underbrace{\frac{v^2}{4} \begin{pmatrix} g_L^2 & -g_L g_Y \\ -g_L g_Y & g_Y^2 \end{pmatrix}}_{\text{mass matrix}} \begin{pmatrix} W^{3,\mu} \\ B^\mu \end{pmatrix} \quad (\otimes)$$

▼ We have the **mass matrix**

$$M^2 \equiv \frac{v^2}{4} \begin{pmatrix} g_L^2 & -g_L g_Y \\ -g_L g_Y & g_Y^2 \end{pmatrix}$$

The mass matrix is **non-diagonal** $\rightarrow$ we cannot read immediately the mass spectrum (W_μ^3 and B_μ are **not** “mass eigenstates”)

First, we need to **diagonalise** the mass matrix. We can do it via an **orthogonal transformation**. We rewrite $(\otimes)$ as

$$\frac{1}{2} (W_\mu^3, B_\mu) O^T (O M^2 O^T) O \begin{pmatrix} W^{3,\mu} \\ B^\mu \end{pmatrix} \quad O \equiv \begin{pmatrix} \cos \theta_W & -\sin \theta_W \\ \sin \theta_W & \cos \theta_W \end{pmatrix}$$

θ_W : “Weinberg angle”

and impose that the transformation O diagonalises M^2 ,

$$O M^2 O^T \stackrel{!}{=} M_\Delta^2, \quad \text{where } M_\Delta^2 \text{ is diagonal.}$$

Added explanation (not in the original notes) — Why an *orthogonal* transformation, and not just any change of basis

This is the technical point that makes the whole procedure legitimate, and the notes state it without comment.

The constraint. The kinetic terms are already canonical: $-\frac{1}{4} W_{\mu\nu}^3 W^{3,\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}$, i.e. proportional to the *identity* in the (W^3, B) space. A general change of basis would spoil that and one would have to renormalise the fields afterwards. An *orthogonal* O ($O^T O = \mathbb{K}$) leaves the identity invariant, so the new fields Z, A have canonical kinetic terms automatically. This is why one may diagonalise the mass matrix and stop.

Why it is possible at all. M^2 is real and symmetric. The spectral theorem guarantees a real orthogonal O diagonalising it with real eigenvalues. And every 2×2 orthogonal matrix with $\det = +1$ is a rotation — hence the single parameter θ_W . The Weinberg angle is not an extra assumption; it is the one free parameter that a 2×2 rotation has.

Reading the insertion trick. The rewriting $\frac{1}{2} \phi^T M^2 \phi = \frac{1}{2} \phi^T O^T (O M^2 O^T) O \phi$ is just inserting $\mathbb{K} = O^T O$ twice. Defining $\phi' \equiv O \phi = (Z, A)^T$, it becomes $\frac{1}{2} \phi'^T M_\Delta^2 \phi'$ — a sum of two independent mass terms. All the content is in finding O .

One historical note. θ_W is variously called the Weinberg angle or the weak mixing angle. It is a *measured* input, not a prediction: $\sin^2 \theta_W \simeq 0.231$. Grand unified theories do predict it (at the unification scale), which is one of their main selling points.

3 Details of the diagonalisation: the eigenvalues

[p. 9*****]

DETAILS OF THE DIAGONALISATION

■ We compute M_{Δ}^2

$$\begin{aligned}\det(M^2 - \lambda \mathbb{K}) &= \det \begin{pmatrix} \frac{v^2 g_L^2}{4} - \lambda & -\frac{v^2 g_L g_Y}{4} \\ -\frac{v^2 g_L g_Y}{4} & \frac{v^2 g_Y^2}{4} - \lambda \end{pmatrix} = \\ &= \left(\frac{v^2 g_L^2}{4} - \lambda \right) \left(\frac{v^2 g_Y^2}{4} - \lambda \right) - \frac{v^4}{16} g_L^2 g_Y^2 = 0 \\ \lambda^2 - \lambda \frac{v^2}{4} (g_L^2 + g_Y^2) + \underbrace{\frac{v^4 g_L^2 g_Y^2}{16} - \frac{v^4 g_L^2 g_Y^2}{16}}_{= 0} &= 0\end{aligned}$$

$$\lambda \left[\lambda - \frac{v^2}{4} (g_L^2 + g_Y^2) \right] = 0$$

$$\rightsquigarrow \begin{cases} \lambda = 0 \\ \lambda = \frac{v^2}{4} (g_L^2 + g_Y^2) \end{cases}$$

eigenvalues

$\Downarrow$

$$M_{\Delta}^2 = \begin{pmatrix} \frac{v^2}{4} (g_L^2 + g_Y^2) & 0 \\ 0 & 0 \end{pmatrix}$$

One gauge boson remains massless!

■ We compute the eigenvectors

$$\frac{v^2}{4} \begin{pmatrix} g_L^2 & -g_L g_Y \\ -g_L g_Y & g_Y^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{v^2}{4} (g_L^2 + g_Y^2) \begin{pmatrix} a \\ b \end{pmatrix}$$

Added explanation (not in the original notes) — the labels (a, b)

The original writes the eigenvector components as (V, W) . Those are just names for two numbers, but V and W are already heavily used for the potential and the gauge fields, so they are renamed (a, b) here. Nothing else changes.

Added explanation (not in the original notes) — Why the determinant vanishes — three ways to see the same thing

The exact cancellation of the two $v^4 g_L^2 g_Y^2 / 16$ terms is the crux of the sheet. It is worth having more than one reason for it.

1. Algebraically. $\det M^2 = \left(\frac{v^2}{4}\right)^2 (g_L^2 g_Y^2 - (g_L g_Y)^2) = 0$. The off-diagonal entry is precisely $\pm$ the geometric mean of the diagonal ones, so the two rows are proportional: $(g_L^2, -g_L g_Y) = g_L \cdot (g_L, -g_Y)$ and $(-g_L g_Y, g_Y^2) = -g_Y \cdot (g_L, -g_Y)$. **The matrix has rank one.** A rank-one 2×2 matrix has eigenvalues $\{0, \text{tr}\}$, which is exactly what the notes find.

2. From the perfect square. M^2 came from $\frac{v^2}{8}(g_Y B - g_L W^3)^2$ — a single squared linear combination. Only *one* direction in field space appears, so only one direction can be massive. The orthogonal direction is untouched.

3. From the symmetry — this is the real reason. Lecture 30 [p. 4*****] showed that $Q = T^3 + Y$ annihilates the vacuum, i.e. a $U(1)$ gauge symmetry survives. By the general theorem of [p. 13*****], the gauge boson of an unbroken generator is *exactly* massless. The vanishing determinant is not an algebraic coincidence to be discovered here; it was guaranteed several sheets ago. If the arithmetic had *not* cancelled, that would have signalled an error upstream.

And the massive eigenvalue is fixed by the trace. Since one eigenvalue is 0 and the trace is basis-independent, $\lambda_{\text{massive}} = \text{tr } M^2 = \frac{v^2}{4}(g_L^2 + g_Y^2)$ — no work needed. The notes get it from the factorised characteristic polynomial, which is the same statement.

4 The eigenvectors and the matrix O

[p. 10*****]

$$\rightsquigarrow \begin{cases} g_L^2 a - g_L g_Y b = (g_L^2 + g_Y^2) a \\ -g_L g_Y a + g_Y^2 b = (g_L^2 + g_Y^2) b \end{cases} \rightsquigarrow \boxed{-g_L b = g_Y a}$$

[p. 10*****]

$$e_1 = \begin{pmatrix} a \\ -\frac{g_Y}{g_L} a \end{pmatrix} \rightarrow \boxed{\hat{e}_1 = \frac{1}{(g_L^2 + g_Y^2)^{1/2}} \begin{pmatrix} g_L \\ -g_Y \end{pmatrix}}$$

$$a^2 \left(1 + \frac{g_Y^2}{g_L^2} \right) = 1 \quad \Rightarrow \quad a = \frac{g_L}{(g_L^2 + g_Y^2)^{1/2}}$$

$$\boxed{\hat{e}_2 = \frac{1}{(g_L^2 + g_Y^2)^{1/2}} \begin{pmatrix} g_Y \\ g_L \end{pmatrix}}$$

■ We construct the matrix O

$$O M^2 O^T = M_\Delta^2 \quad \text{with}$$

$$\boxed{O = \frac{1}{(g_L^2 + g_Y^2)^{1/2}} \begin{pmatrix} g_L & -g_Y \\ g_Y & g_L \end{pmatrix}}$$

which gives the **mass eigenstates**

$$\boxed{O \begin{pmatrix} W^{3,\mu} \\ B^\mu \end{pmatrix} = \frac{1}{(g_L^2 + g_Y^2)^{1/2}} \begin{pmatrix} g_L & -g_Y \\ g_Y & g_L \end{pmatrix} \begin{pmatrix} W^{3,\mu} \\ B^\mu \end{pmatrix} = \begin{pmatrix} Z^\mu \\ A^\mu \end{pmatrix}}$$

with the identifications

Added explanation (not in the original notes) — Checking the eigenvector equation, and why the rows of O are the eigenvectors

The first equation, solved. $g_L^2 a - g_L g_Y b = (g_L^2 + g_Y^2) a$ gives $-g_L g_Y b = g_Y^2 a$, i.e. $-g_L b = g_Y a$ after dividing by g_Y . The second equation gives the same relation — as it must, since the matrix is singular and the two equations are proportional. So $b = -\frac{g_Y}{g_L} a$, and normalising $a^2 + b^2 = a^2 (1 + g_Y^2/g_L^2) = 1$ gives $a = g_L / (g_L^2 + g_Y^2)^{1/2}$.

Where $\hat{e}_2$ comes from. The notes simply write it down. In two dimensions the second eigenvector of a symmetric matrix is forced to be orthogonal to the first, and the unique unit vector orthogonal to $(g_L, -g_Y)$ is (g_Y, g_L) (up to sign). One can also check directly that $M^2(g_Y, g_L)^T = 0$: the first row gives $g_L^2 g_Y - g_L g_Y g_L = 0 \checkmark$. So $\hat{e}_2$ is the *massless* direction.

Why O has the eigenvectors as rows. The defining relation is $OM^2O^T = M_\Delta^2$, i.e. $(OM^2O^T)_{ij} = \hat{e}_i \cdot M^2 \hat{e}_j = \lambda_j \delta_{ij}$ when the i -th row of O is $\hat{e}_i$. And $(O\phi)_i = \hat{e}_i \cdot \phi$, so the i -th new field is the projection of the old fields onto the i -th eigenvector. Hence

$$Z^\mu = \hat{e}_1 \cdot (W^3, B) = \frac{g_L W^{3,\mu} - g_Y B^\mu}{(g_L^2 + g_Y^2)^{1/2}}, \quad A^\mu = \hat{e}_2 \cdot (W^3, B) = \frac{g_Y W^{3,\mu} + g_L B^\mu}{(g_L^2 + g_Y^2)^{1/2}}.$$

Z picks up the massive eigenvalue because $\hat{e}_1$ was the eigenvector with $\lambda = \frac{v^2}{4}(g_L^2 + g_Y^2)$; A is massless. Which one is called Z is decided by the row order of M_Δ^2 chosen on [p. 9*****] — massive first.

Consistency with O as a rotation. On [p. 8*****], O was posited as $\begin{pmatrix} \cos \theta_W & -\sin \theta_W \\ \sin \theta_W & \cos \theta_W \end{pmatrix}$.

Comparing with the explicit O just constructed forces $\cos \theta_W = g_L/(g_L^2 + g_Y^2)^{1/2}$ and $\sin \theta_W = g_Y/(g_L^2 + g_Y^2)^{1/2}$ — which is precisely the identification the next sheet makes. Note $\det O = (g_L^2 + g_Y^2)/(g_L^2 + g_Y^2) = +1$ ✓: a genuine rotation, no reflection.

5 The Weinberg angle, Z and A , and the final kinetic term

[p. 11*****]

$$\sin \theta_W = \frac{g_Y}{(g_L^2 + g_Y^2)^{1/2}}$$

$$\cos \theta_W = \frac{g_L}{(g_L^2 + g_Y^2)^{1/2}}$$

[p. 11*****]

We have

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu$$

$$A_\mu = \cos \theta_W B_\mu + \sin \theta_W W_\mu^3$$

► massive eigenstate

► massless eigenstate

so that

$$\frac{v^2}{8} (g_Y B_\mu - g_L W_\mu^3)^2 = \frac{1}{2} M_Z^2 Z_\mu Z^\mu ; \quad M_Z^2 = \frac{v^2}{4} (g_L^2 + g_Y^2)$$

↘ from which we also have

$$(g_Y B_\mu - g_L W_\mu^3)^2 = (g_L^2 + g_Y^2) Z_\mu Z^\mu$$

The covariant Higgs kinetic term, therefore, gives

$$\begin{aligned} (D_\mu H)^\dagger (D^\mu H) &= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \underbrace{\frac{1}{2} M_W^2 [(W_\mu^1)^2 + (W_\mu^2)^2] + \frac{1}{2} M_Z^2 Z_\mu Z^\mu}_{\text{mass terms for the gauge bosons } W^{1,2} \text{ and } Z} \\ &\quad + \frac{g_L^2}{8} (h^2 + 2hv) [(W_\mu^1)^2 + (W_\mu^2)^2] + \frac{1}{8} (g_L^2 + g_Y^2) (h^2 + 2hv) Z_\mu Z^\mu \end{aligned}$$

Added explanation (not in the original notes) — Reading the two identifications, and a sanity check on the whole page

Where $\sin \theta_W$ and $\cos \theta_W$ come from. Matching the explicit O of [p. 10*****] against the rotation form of [p. 8*****] forces $\cos \theta_W = g_L / (g_L^2 + g_Y^2)^{1/2}$, $\sin \theta_W = g_Y / (g_L^2 + g_Y^2)^{1/2}$. Note $\sin^2 + \cos^2 = (g_L^2 + g_Y^2) / (g_L^2 + g_Y^2) = 1$ ✓, and

$$\tan \theta_W = \frac{g_Y}{g_L}$$

so the mixing angle measures the *ratio* of the two gauge couplings. If $g_Y \rightarrow 0$ the angle vanishes, $Z \rightarrow W^3$ and $A \rightarrow B$: no mixing, and the $U(1)_Y$ boson would be the photon. The mixing is entirely due to hypercharge being switched on.

The rewriting into Z , step by step. From [p. 10*****], $Z_\mu = (g_L W_\mu^3 - g_Y B_\mu) / (g_L^2 + g_Y^2)^{1/2}$, so

$$(g_Y B_\mu - g_L W_\mu^3)^2 = \left(- (g_L^2 + g_Y^2)^{1/2} Z_\mu \right)^2 = (g_L^2 + g_Y^2) Z_\mu Z^\mu,$$

which is the green line. Multiplying by $v^2/8$: $\frac{v^2}{8}(g_L^2 + g_Y^2)Z_\mu Z^\mu = \frac{1}{2}\left[\frac{v^2}{4}(g_L^2 + g_Y^2)\right]Z_\mu Z^\mu$, hence $M_Z^2 = \frac{v^2}{4}(g_L^2 + g_Y^2) \checkmark$ — matching the eigenvalue found on [p. 9*****], as it must.

The ρ relation falls out immediately. Divide the two masses:

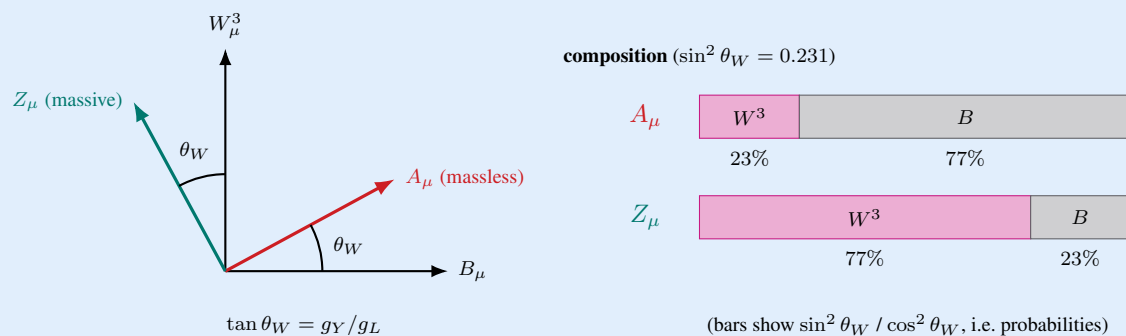
$$\frac{M_W^2}{M_Z^2} = \frac{\frac{1}{4}g_L^2 v^2}{\frac{1}{4}(g_L^2 + g_Y^2)v^2} = \frac{g_L^2}{g_L^2 + g_Y^2} = \cos^2 \theta_W \implies \boxed{M_Z = \frac{M_W}{\cos \theta_W}}$$

This is the $\rho = 1$ prediction of Lecture 30, now derived rather than asserted. It is a non-trivial consequence of the Higgs being an $SU(2)_L$ *doublet*; a scalar in another representation would give $\rho \neq 1$ at tree level. Experimentally $\rho = 1.00038 \pm 0.00020$.

Numerically. With $\sin^2 \theta_W = 0.231$: $M_Z = M_W / \cos \theta_W = 80.4 / 0.877 = 91.7$ GeV against the measured 91.19 GeV — the residual per-cent-level difference is radiative corrections, which is exactly the level at which they should appear.

What the last line of the box is for. The $(h^2 + 2hv)$ terms are *not* masses; they are the interactions between the physical Higgs and the massive gauge bosons. They are rewritten in terms of M_W and M_Z on [p. 16*****], and they are the couplings the LHC measures.

Added explanation (not in the original notes) — Figure — the rotation, and what each mass eigenstate is made of



The photon is *mostly* the hypercharge boson B ; the Z is *mostly* W^3 . Neither is purely one or the other — which is why the photon couples to a charge $Q = T^3 + Y$ mixing both quantum numbers, and why the electromagnetic coupling $e = g_L \sin \theta_W = g_Y \cos \theta_W$ is smaller than either g_L or g_Y .

Note the asymmetry of the picture: because $g_L > g_Y$ (i.e. $\theta_W < 45^\circ$), the rotation is “mostly the identity” — the fields do not swap roles, they tilt.

PART II — The unbroken $U(1)$ and electric charge [p. 12*****]–[p. 14*****]

6 The photon, and the Higgs field

[p. 12*****]

The gauge boson A_μ remains massless.

↪ We would like to interpret A_μ as the **ordinary massless photon** and, consequently, the unbroken $U(1)$ as the gauge group of electromagnetism [p. 12*****]

Before proceeding, let us therefore explore the **transformation properties of the fields** under the unbroken $U(1)$.

We should investigate how the fields transform under the $U(1)$ symmetry generated by Q .

We can do it **almost without computations**:

■ THE HIGGS FIELD

Since $H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$,

$h(x)$ inherits the same transformation properties of the vacuum under the unbroken $U(1)$

$$h(x) \xrightarrow[U(1)]{\text{unbroken}} h(x)$$

Since we want to identify $U(1)$ with the $U(1)$ of electromagnetism, we say that the Higgs field is electrically neutral

Added explanation (not in the original notes) — “Almost without computations” — what the argument actually is

The reasoning on this sheet is slick and easy to read past. Spelled out:

The one-line argument. By construction $QH_0 = 0$: the vacuum is annihilated by Q , i.e. $e^{-i\theta Q}H_0 = H_0$. In unitary gauge the *whole field* is $H(x) = H_0 \cdot \frac{v+h(x)}{v}$ — it points in the same direction in $SU(2) \times U(1)$ space as the vacuum, with only its length fluctuating. A transformation that leaves the direction alone therefore leaves the fluctuation alone too. Hence $h \rightarrow h$: charge zero.

Why this is a consistency requirement, not a choice. If h carried electric charge, then $\langle H \rangle \neq 0$ would break electromagnetism and the photon would be massive. The statement “the Higgs is neutral” and the statement “the photon is massless” are the same statement, viewed from two sides. This is also why the vev had to sit in the *lower* component of the doublet: that is the component with $Q = T^3 + Y = -\frac{1}{2} + \frac{1}{2} = 0$.

The pattern for the rest of Part II. Every field is treated the same way: ask how $e^{-i\theta Q}$ acts on it, read off the phase, call the exponent the charge. No Lagrangian, no Feynman rules — pure representation theory. Three fields, three answers:

field	transformation	charge
h	$h \rightarrow h$	0 ([p. 12*****])
$W_\mu^\pm$	$W_\mu^\pm \rightarrow e^{\mp i\theta} W_\mu^\pm$	± 1 ([p. 13*****])
Z_μ	$Z_\mu \rightarrow Z_\mu$	0 ([p. 14*****])

And A_μ itself is neutral — the photon does not carry electric charge, which is precisely what makes electromagnetism abelian and why photons do not scatter off each other at tree level.

7 The massive gauge bosons $W^{1,2}$

[p. 13*****]

THE MASSIVE GAUGE BOSONS $W^{1,2}$

$W_\mu^{1,2,3}$ transform according to the **adjoint** of $SU(2)_L$ and **do not transform** under $U(1)_Y$.

[p. 13*****]

Consequently,

$$Q = T^3 + Y = t^3$$

This is the same situation we studied in the Georgi–Glashow model

We thus write

$$W_\mu^3 \xrightarrow[U(1)]{\text{unbroken}} W_\mu^3$$

$$W_\mu^+ \xrightarrow[U(1)]{\text{unbroken}} e^{-i\theta} W_\mu^+$$

electric charge +1

$$W_\mu^- \xrightarrow[U(1)]{\text{unbroken}} e^{+i\theta} W_\mu^-$$

electric charge −1

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2)$$

$W_\mu^\pm$ create/destroy massive spin-1 particles/antiparticles with mass squared M_W^2 and charges ± 1

Added explanation (not in the original notes) — Why Q collapses to t^3 on the gauge bosons — and why that is the whole point

The argument. Electric charge is $Q = T^3 + Y$, where T^3 and Y mean “the generators *in the representation the field lives in*”. Gauge fields live in the adjoint of their own group:

- W_μ^A sits in the adjoint of $SU(2)_L$, so T^3 acts on it as the adjoint generator t^3 , with $(t^3)_{BC} = -i\varepsilon_{3BC}$;
- W_μ^A carries *no hypercharge* — B_μ is a different, commuting gauge field and $[SU(2)_L, U(1)_Y] = 0$ — so $Y = 0$ on it.

Hence $Q = t^3 + 0 = t^3$. The electric charge operator, restricted to the weak gauge bosons, is exactly the generator whose eigenvalue problem was solved in the Georgi–Glashow model. That is why the magenta remark says “the same situation”.

Recycling the Georgi–Glashow result verbatim. There (Lecture 30 [p. 27*****]–[p. 29*****]), $e^{-i\theta t^3}$ acting on the adjoint triplet gave: W^3 inert, (W^1, W^2) rotating into each other, and the combinations diagonalising the rotation being $W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2)$ with eigenvalues $e^{\mp i\theta}$. Nothing has to be recomputed. **This is the payoff of having done that model:** the entire charge structure of the W bosons is transferred wholesale.

Reading off the charge from the phase. A field of charge q transforms as $\psi \rightarrow e^{-iq\theta}\psi$. Comparing with $W^+ \rightarrow e^{-i\theta}W^+$ gives $q = +1$, and $W^- \rightarrow e^{+i\theta}W^-$ gives $q = -1$. Note that $W^- = (W^+)^\dagger$, so the two are antiparticles of each other — consistent with opposite charges.

A subtlety worth stating. W_μ^3 is inert under Q , and so are B_μ , Z_μ and A_μ — all four neutral fields. But W^3 is *not* a mass eigenstate, so it is not a particle; the physical neutral vectors are Z and γ . Charge assignment and mass diagonalisation are independent questions, and they happen to be compatible here only because Q commutes with the mass matrix.

8 The mode expansion of $W^\pm$, and the Z

[p. 14*****]

To be even more precise, if we write

[p. 14*****]

$$W_\mu^+ = \sum_{\lambda=0,\pm 1} \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3} 2E_k} \left[\underbrace{a(\vec{p}, \lambda, +1) \varepsilon_\mu(\vec{p}, \lambda, +1) e^{-ip \cdot x}}_{(1)} + \underbrace{b^\dagger(\vec{p}, \lambda, -1) \varepsilon_\mu(\vec{p}, \lambda, -1)^* e^{ip \cdot x}}_{(2)} \right]$$

(1) **destroy** particles with charge +1, which we call “ W^+ ”

(2) **create** anti-particles with charge -1 , which we call “ W^- ”

$$W_\mu^- = (W_\mu^+)^\dagger = \sum_{\lambda=0,\pm 1} \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3} 2E_k} \left[\underbrace{a^\dagger(\vec{p}, \lambda, +1) \varepsilon_\mu(\vec{p}, \lambda, +1)^* e^{ip \cdot x}}_{(3)} + \underbrace{b(\vec{p}, \lambda, -1) \varepsilon_\mu(\vec{p}, \lambda, -1) e^{-ip \cdot x}}_{(4)} \right]$$

(3) **create** “ W^+ ”

(4) **destroy** “ W^- ”

■ THE MASSIVE GAUGE BOSON Z

As we have seen, W_μ^3 does not transform. Since B_μ is an abelian gauge boson, it does not transform under the global symmetry. Consequently,

$$Z_\mu \xrightarrow[U(1)]{\text{unbroken}} Z_\mu$$

electrically neutral

Added explanation (not in the original notes) — What the third label on the operators is doing — and the three sums

The three labels. $a(\vec{p}, \lambda, +1)$ carries momentum, helicity, *and charge*. The first two are familiar from any massive vector field; the third is the new content of this sheet. It records that the quanta of W_μ^+ are eigenstates of Q — which is exactly what [p. 13*****] established, now expressed at the level of the mode operators.

$\lambda = 0, \pm 1$: **three polarisations, not two.** The sum runs over three helicity states because $W^\pm$ is *massive*. The $\lambda = 0$ state is the longitudinal polarisation — and by Lectures 29–30 it *is* the eaten Goldstone. So the third term in this sum is literally where one of the three Higgs-doublet components went.

Why $a \neq b$ — i.e. why $W^+ \neq W^-$. For a *real* field one has $b = a$ and the particle is its own antiparticle (the photon, the Z). Here W_μ^+ is *complex*, so a and b are independent operators and the quanta come in particle/antiparticle pairs of opposite charge. *A field can carry a conserved charge if and only if it is complex.* This is the field-theoretic version of the statement on [p. 13*****] that one must complexify $W^{1,2}$ to speak of charge at all.

The bookkeeping, checked. W_μ^+ must *lower* the electric charge of a state by one unit (it appears in the Lagrangian contracted with a current that raises it). It does so in two ways: destroying a W^+ ,

or creating a W^- . Both change Q by -1 . ✓ $W_\mu^- = (W_\mu^+)^\dagger$ does the reverse. This is exactly the structure of the complex scalar field, one spin higher.

Why Z is neutral, in one line. $Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu$, and both W^3 and B are Q -inert — W^3 because $\varepsilon_{333} = 0$, and B because an abelian gauge field never transforms under its own group (or any other, here). A linear combination of two inert fields is inert. Consequently Z_μ is real, $Z_\mu = Z_\mu^\dagger$, and the Z boson is its own antiparticle.

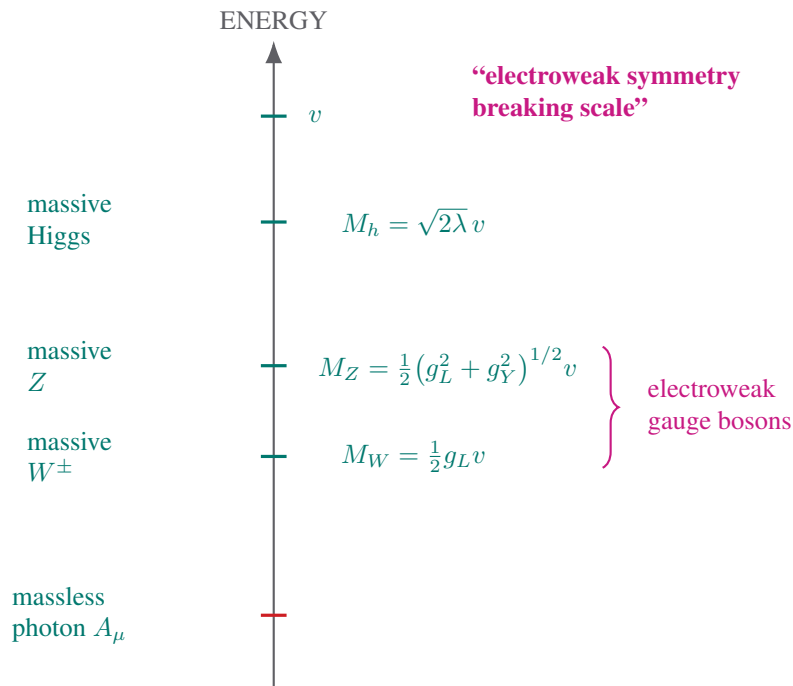
PART III — The spectrum and the bosonic interactions [p. 15***]–[p. 18*****]**

9 The mass spectrum

[p. 15*****]

The mass spectrum is

[p. 15*****]


Added explanation (not in the original notes) — Reading the ladder — what sets each rung, and the measured numbers

Every entry is v times a coupling. That is the whole content of the picture: *one scale, several dimensionless numbers*.

state	formula	coupling	measured
Higgs h	$M_h = \sqrt{2\lambda} v$	$\lambda \simeq 0.13$	125.25 GeV
Z	$M_Z = \frac{1}{2} \sqrt{g_L^2 + g_Y^2} v = M_W / \cos \theta_W$	g_L, g_Y	91.19 GeV
$W^\pm$	$M_W = \frac{1}{2} g_L v$	$g_L \simeq 0.65$	80.38 GeV
photon	0	—	$< 10^{-18}$ eV

Why the ordering comes out this way. $M_Z > M_W$ always, since $M_Z = M_W / \cos \theta_W$ and $\cos \theta_W < 1$. And $M_h > M_Z$ requires $\sqrt{2\lambda} > \frac{1}{2} \sqrt{g_L^2 + g_Y^2}$, which happens to hold — but is not forced by the theory. λ is a free parameter, so the Higgs could in principle have sat anywhere on this axis. Where it sits was measured, not predicted.

What “electroweak symmetry breaking scale” means. $v = 246$ GeV is not a mass of anything; it is the vev, the size of the condensate. Every mass in the table is v times something smaller than one, so the whole spectrum lies *below* v . That the couplings are all $\mathcal{O}(0.1\text{--}1)$ is why the electroweak sector is weakly coupled and perturbation theory works.

And v itself is measured how? From the muon lifetime: $G_F/\sqrt{2} = 1/(2v^2)$, giving $v = 246.22$ GeV. So this scale was known accurately decades before any of the particles on the ladder were found.

The photon's masslessness is not approximate. It is protected by the unbroken $U(1)_{\text{em}}$ gauge symmetry, exactly, to all orders — see the theorem of Lecture 30 [p. 13^{*****}]. The experimental bound in the table is a bound on a symmetry-violating parameter, not on a small calculable quantity.

10 Interactions in the bosonic sector — three categories

[p. 15*****]–[p. 16*****]

INTERACTIONS IN THE BOSONIC SECTOR

Three categories:

1) Higgs self-interactions

$$\mathcal{L}_{\text{int}} = -\lambda v h^3 - \frac{\lambda}{4} h^4$$



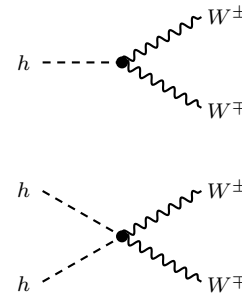
2) Interactions of the Higgs with the gauge bosons

[p. 16*****]

$$\frac{g_L^2}{8} (h^2 + 2hv) \underbrace{[(W_\mu^1)^2 + (W_\mu^2)^2]}_{= 2W_\mu^+ W^{-,\mu}} = \frac{1}{2} \left(\frac{M_W^2}{v^2} \right) (h^2 + 2hv) 2W_\mu^+ W^{-,\mu}$$

$$\uparrow g_L^2 = 4M_W^2/v^2$$

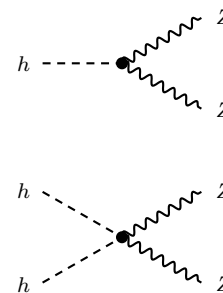
$$\mathcal{L}_{\text{int}} = \frac{M_W^2}{v^2} (h^2 + 2hv) W_\mu^+ W^{-,\mu}$$



$$\frac{1}{8} (g_L^2 + g_Y^2) (h^2 + 2hv) Z_\mu Z^\mu = \frac{1}{2} \left(\frac{M_Z^2}{v^2} \right) (h^2 + 2hv) Z_\mu Z^\mu$$

$$\hookrightarrow (g_L^2 + g_Y^2) = 4M_Z^2/v^2$$

$$\mathcal{L}_{\text{int}} = \frac{M_Z^2}{2v^2} (h^2 + 2hv) Z_\mu Z^\mu$$



Added explanation (not in the original notes) — Verifying $[(W^1)^2 + (W^2)^2] = 2W_\mu^+ W^{-,\mu}$, and the significance of the rewriting

The identity. With $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2)$,

$$2W_\mu^+ W^{-,\mu} = 2 \cdot \frac{1}{2} (W_\mu^1 - iW_\mu^2) (W^{1,\mu} + iW^{2,\mu}) = W^1 W^1 + W^2 W^2 + \underbrace{i(W^1 W^2 - W^2 W^1)}_{=0} \cdot \checkmark$$

The cross terms cancel because the fields commute. (Note $W^\pm$ is complex, so $W^+ W^-$ — not

W^+W^+ — is the real, Lorentz-invariant bilinear.)

Why rewrite the couplings in terms of M_W and M_Z . Substituting $g_L^2 = 4M_W^2/v^2$ and $(g_L^2 + g_Y^2) = 4M_Z^2/v^2$ turns the couplings into

$$\mathcal{L} \supset \frac{2M_W^2}{v} h W_\mu^+ W^{-,\mu} + \frac{M_Z^2}{v} h Z_\mu Z^\mu.$$

This is the sharp, testable statement of the mechanism: **the Higgs couples to each massive vector in proportion to that vector's mass squared, divided by v** , with no free parameter. The Higgs does not couple to the photon at tree level at all — consistent with $M_\gamma = 0$, and the reason $h \rightarrow \gamma\gamma$ proceeds only through loops.

The relative factor of 2 between W and Z . W^+W^- involves two *distinct* fields while ZZ involves the same field twice, so the ZZ term carries the usual $\frac{1}{2}$ symmetry factor. Writing both as $\frac{2M_V^2}{v} h V_\mu V^\mu \times S_V$ with $S_W = 1$, $S_Z = \frac{1}{2}$ makes the pattern uniform. This is bookkeeping, not physics.

The $hhVV$ vertices. The h^2 pieces give four-point couplings $\frac{M_W^2}{v^2} h^2 W^+W^-$ and $\frac{M_Z^2}{2v^2} h^2 ZZ$ — again fixed, with no new parameters. They are needed for the unitarity cancellation of Lecture 30 [p. 21*****] and are being probed at the LHC through double-Higgs and vector-boson-scattering processes.

11 Self-interactions among gauge bosons

[p. 17*****]–[p. 18*****]

3) Self-interactions among gauge bosons

The pure Yang–Mills Lagrangian

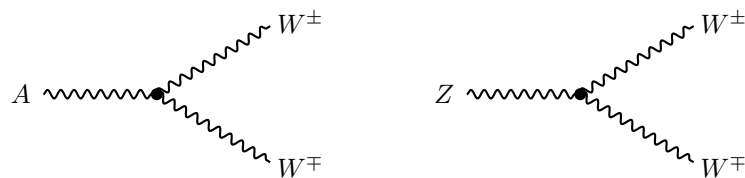
$$-\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}$$

gives rise to interactions among the electroweak gauge bosons and the photon

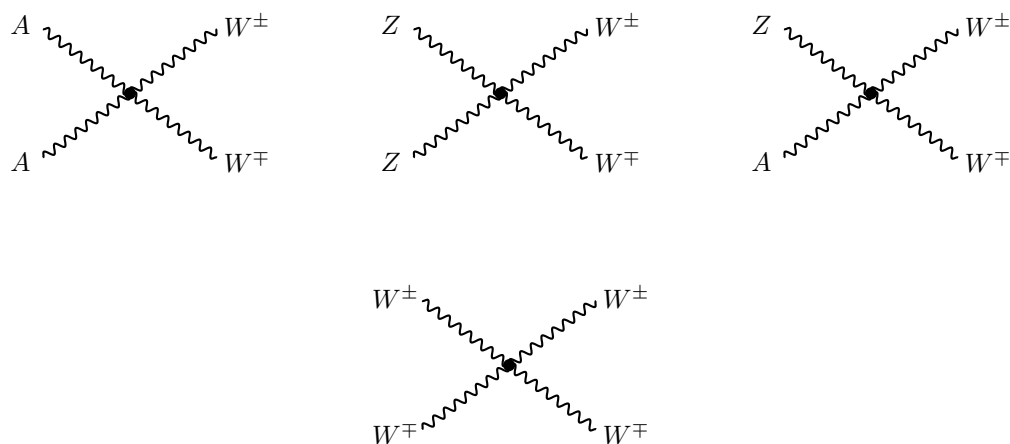
↪ To obtain $\mathcal{L}_{\text{int}}$, we should write the interaction pieces of the Lagrangian in the **mass eigenstates** $W_\mu^\pm, Z_\mu, A_\mu$

Without doing computations, we can **list the possible interactions** considering that they must respect electric charge conservation

■ TRI-LINEAR VERTICES (cubic)



■ QUARTIC SELF-INTERACTIONS



Added explanation (not in the original notes) — Charge conservation as a complete selection rule — checking the list

The green remark is doing real work: *charge conservation alone determines which vertices can exist*. Every vertex must have $\sum Q = 0$, and the available charges are $Q(W^\pm) = \pm 1$, $Q(Z) = Q(A) = 0$.

Cubic vertices. With three legs, the options are:

- W^+W^-A and W^+W^-Z — charges $+1 - 1 + 0 = 0$ ✓ **allowed**;
- ZZZ, ZZA, ZAA, AAA — charge-neutral, but *absent*. Why? Because the Yang–Mills interactions come from $-\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu}$ with structure constants ε_{ABC} , which is totally *antisymmetric*: a vertex needs three *different* $SU(2)$ indices. Any all-neutral vertex would need W^3 (or B) three times. So charge conservation is necessary but the group structure is what kills these.
- $W^+W^+W^-$ etc. — charge $+1$, **forbidden**.

Quartic vertices. Charge-neutral combinations with four legs, at most two of which can be charged (since ε_{ABC} appears squared, giving at most two $SU(2)$ index pairs): W^+W^-AA , W^+W^-ZZ , W^+W^-ZA , $W^+W^-W^+W^-$. That is *exactly* the list drawn on [p. 17*****]–[p. 18*****] — four diagrams, no more.

One important absence: no $\gamma\gamma\gamma\gamma$, no $ZZZZ$, no $\gamma\gamma\gamma$. The photon has no tree-level self-interaction, which is the statement that QED is abelian. Light-by-light scattering exists but only at one loop, through a charged particle running in the loop.

Where the couplings come from. Writing $W^3 = \cos\theta_W Z + \sin\theta_W A$ inside $\varepsilon_{ABC}W_\mu^A W_\nu^B$ produces the two cubic vertices with strengths

$$g_{WWA} = g_L \sin\theta_W = e, \quad g_{WWZ} = g_L \cos\theta_W.$$

The first is a *prediction*: the photon couples to the $W^\pm$ with exactly the electric charge e , as it must, since $W^\pm$ has $Q = \pm 1$. *This is the check that the identification of A_μ with the photon is consistent* — the notes assert the identification on [p. 12*****], and this is what justifies it.

12 The fermionic sector — opening

[p. 18*****]

THE FERMIONIC SECTOR

Preliminary consideration. Consider a **Dirac fermion**

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad \begin{array}{ll} \blacktriangleright \psi_L \sim \left(\frac{1}{2}, 0\right) & \text{left-handed Weyl spinor} \\ \blacktriangleright \psi_R \sim \left(0, \frac{1}{2}\right) & \text{right-handed Weyl spinor} \end{array}$$

We write, equivalently:

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \psi_R \end{pmatrix} \equiv \Psi_L + \Psi_R = \underbrace{P_L \Psi} + \underbrace{P_R \Psi}$$

$$P_L = \frac{1}{2}(1 - \gamma^5) \quad P_R = \frac{1}{2}(1 + \gamma^5)$$

The Standard Model is a chiral theory because, for each fermion, the left and right components transform differently under $SU(3)_c \times SU(2)_L \times U(1)_Y$

[the scan ends here]

Added explanation (not in the original notes) — The projectors, and why “chiral” is about to change everything

Check that $P_{L,R}$ are projectors. Using $(\gamma^5)^2 = \mathbb{1}$:

$$P_L^2 = \frac{1}{4}(1 - \gamma^5)^2 = \frac{1}{4}(1 - 2\gamma^5 + 1) = \frac{1}{2}(1 - \gamma^5) = P_L,$$

and similarly $P_R^2 = P_R$, $P_L P_R = \frac{1}{4}(1 - \gamma^5)(1 + \gamma^5) = \frac{1}{4}(1 - 1) = 0$, and $P_L + P_R = \mathbb{1}$. So they are complete, orthogonal projectors: any Dirac spinor splits uniquely into two Weyl pieces, and the split is Lorentz invariant (because γ^5 commutes with the Lorentz generators $\Sigma^{\mu\nu}$).

The Lorentz labels. $\psi_L \sim (\frac{1}{2}, 0)$ and $\psi_R \sim (0, \frac{1}{2})$ are the two inequivalent two-dimensional irreps of $\mathfrak{sl}(2, \mathbb{C})$. They are exchanged by parity — which is why a theory that treats them differently *violates parity*, as the weak interaction famously does.

Why the last sentence is the hinge of the whole Standard Model. If ψ_L and ψ_R transform in *different* representations of the gauge group, then the Dirac mass term

$$m \bar{\Psi} \Psi = m(\bar{\Psi}_L \Psi_R + \bar{\Psi}_R \Psi_L)$$

is **not gauge invariant** — it pairs objects in different representations. So *every fermion in the Standard Model is forbidden a mass by gauge symmetry*. The only way out is to soak up the mismatch with the Higgs doublet, via a Yukawa coupling $y \bar{\Psi}_L H \Psi_R$, which becomes a mass $yv/\sqrt{2}$ only after the symmetry breaks. That is why the Higgs is responsible for *all* the masses in the theory, not just the gauge bosons’.

The scan stops exactly at that observation. Section 13 carries it forward.

13 Where this is going — the fermionic sector

Added explanation (not in the original notes) — Everything in this section is added; it is not in the original notes

The scan stops one sentence into the fermionic sector, with the projectors defined and the word “chiral” explained but not yet used. What follows sketches where the next lecture must go. It is not a substitute for that lecture — it is the shape of the argument, so that the last sheet does not read as a fragment.

Step 1 — the representation table

“The left and right components transform differently” is a statement that needs numbers. Here they are, for one generation. Recall the labelling $(SU(3)_c, SU(2)_L, U(1)_Y)$ of Lecture 30 [p. 2*****], and that $Q = T^3 + Y$.

field	content	rep	Y	$Q = T^3 + Y$
Q_L	$\begin{pmatrix} u_L \\ d_L \end{pmatrix}$	$(\mathbf{3}, \mathbf{2}, \frac{1}{6})$	$\frac{1}{6}$	$+\frac{2}{3}, -\frac{1}{3}$
u_R	u_R	$(\mathbf{3}, \mathbf{1}, \frac{2}{3})$	$\frac{2}{3}$	$+\frac{2}{3}$
d_R	d_R	$(\mathbf{3}, \mathbf{1}, -\frac{1}{3})$	$-\frac{1}{3}$	$-\frac{1}{3}$
L_L	$\begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$	$(\mathbf{1}, \mathbf{2}, -\frac{1}{2})$	$-\frac{1}{2}$	$0, -1$
e_R	e_R	$(\mathbf{1}, \mathbf{1}, -1)$	-1	-1
H	$\begin{pmatrix} H^+ \\ H^0 \end{pmatrix}$	$(\mathbf{1}, \mathbf{2}, \frac{1}{2})$	$\frac{1}{2}$	$+1, 0$

Read the second column against the third. The left-handed fields are $SU(2)_L$ *doublets*; the right-handed ones are *singlets*. That is the asymmetry the word “chiral” names, and it is why only left-handed fermions feel the $W^\pm$. Note also that every charge in the last column comes out right — $+\frac{2}{3}$ for the up quark, -1 for the electron, 0 for the neutrino — from hypercharge assignments chosen once and never adjusted again.

Step 2 — fermion masses are forbidden, and then restored

A Dirac mass pairs Ψ_L with Ψ_R . Check the quantum numbers of $\bar{e}_L e_R$: it is an $SU(2)_L$ doublet (from $\bar{e}_L$) with hypercharge $+\frac{1}{2} - 1 = -\frac{1}{2}$. **Not a singlet: not gauge invariant.** The same for every fermion. So the naive mass term is illegal.

The Higgs doublet supplies exactly the missing quantum numbers. The combination

$$\mathcal{L}_{\text{Yuk}} = -y_e \bar{L}_L H e_R - y_d \bar{Q}_L H d_R - y_u \bar{Q}_L \tilde{H} u_R + \text{h.c.}, \quad \tilde{H} \equiv i\sigma^2 H^*,$$

is invariant: $\bar{L}_L H$ is $\mathbf{2} \otimes \mathbf{2} \supset \mathbf{1}$ with $Y = +\frac{1}{2} + \frac{1}{2} = 1$, matching $Y(e_R) = -1$. After symmetry breaking, $H \rightarrow \frac{1}{\sqrt{2}}(0, v + h)^T$ and

$$\mathcal{L}_{\text{Yuk}} \longrightarrow - \underbrace{\frac{y_f v}{\sqrt{2}}}_{m_f} \bar{\Psi}_f \Psi_f - \frac{m_f}{v} h \bar{\Psi}_f \Psi_f.$$

Two things at once: a mass $m_f = y_f v / \sqrt{2}$, and a coupling of the Higgs to that fermion of strength m_f / v — again *proportional to the mass*, exactly as for the gauge bosons on [p. 16*****]. This is the

single most tested prediction of the Higgs sector, and it is why the LHC sees $h \rightarrow b\bar{b}$ and $h \rightarrow \tau\tau$ but not $h \rightarrow ee$.

Added explanation (not in the original notes) — Why $\tilde{H}$ is needed, and what the Yukawa couplings do *not* explain

The conjugate doublet. H has $Y = +\frac{1}{2}$, which is what $\bar{Q}_L d_R$ needs ($-\frac{1}{6} - \frac{1}{3}$ requires $+\frac{1}{2}$). But $\bar{Q}_L u_R$ needs $-\frac{1}{2}$. In a general gauge theory one would need a second Higgs doublet; in $SU(2)$ one gets away with $\tilde{H} = i\sigma^2 H^*$, which is again a doublet (because **2** is pseudo-real) but with $Y = -\frac{1}{2}$. This is a peculiarity of $SU(2)$ and it is why the Standard Model needs only *one* Higgs doublet. Two-Higgs-doublet models — including supersymmetry — give this up.

What is explained and what is not. The mechanism explains why fermions *can* have masses at all despite the chiral gauge symmetry, and it predicts that the Higgs coupling to each fermion is m_f/v . It does *not* explain the values: $y_t \simeq 1$ while $y_e \simeq 3 \times 10^{-6}$, a spread of six orders of magnitude, all put in by hand. That is the *flavour puzzle*, and it is the largest unexplained structure in the Standard Model.

And the CKM matrix. With three generations the Yukawas are 3×3 matrices. Diagonalising them rotates the quark fields, and the mismatch between the up- and down-type rotations survives in the charged current as the CKM matrix — the origin of quark mixing and of CP violation. None of this exists in the one-generation sketch above.

Step 3 — the currents, and why the theory hangs together

Expanding $\bar{\Psi} i \gamma^\mu D_\mu \Psi$ in the mass eigenstates produces three currents:

charged current:	$\frac{gL}{\sqrt{2}} W_\mu^+ \bar{u}_L \gamma^\mu d_L + \text{h.c.}$	<i>left-handed only</i> — parity violation
electromagnetic:	$e A_\mu \bar{\Psi} \gamma^\mu Q \Psi$	<i>vector-like</i> — parity conserving
neutral current:	$\frac{gL}{\cos \theta_W} Z_\mu \bar{\Psi} \gamma^\mu (T^3 - Q \sin^2 \theta_W) \Psi$	<i>mixed</i> — the Z sees both

The middle line is the consistency check that closes the whole construction: because $A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu$ couples through $Q = T^3 + Y$, and Q is *the same* for left- and right-handed components of a given fermion, the photon coupling is automatically parity-conserving and equal to e times the electric charge. Ordinary QED emerges, unmodified, out of a manifestly chiral theory.

Added explanation (not in the original notes) — Anomaly cancellation — the constraint that makes a chiral theory legal

A chiral gauge theory is not automatically consistent. Triangle diagrams with three gauge bosons and a chiral fermion loop can generate a *gauge anomaly*, which destroys gauge invariance at one loop and with it renormalisability and unitarity. The anomaly is proportional to a sum over all left-handed Weyl fermions, and it must vanish identically.

Writing everything as left-handed ($u_R \rightarrow u_R^c$ with $Y \rightarrow -Y$, etc.), for one generation:

condition	sum	value
$[SU(3)]^2 U(1)_Y$	$2(\frac{1}{6}) + (-\frac{2}{3}) + (\frac{1}{3})$	0 ✓
$[SU(2)]^2 U(1)_Y$	$3(\frac{1}{6}) + 1(-\frac{1}{2})$	0 ✓
$[U(1)_Y]^3$	$6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 + (1)^3$	0 ✓
$[\text{grav}]^2 U(1)_Y$	$6(\frac{1}{6}) + 3(-\frac{2}{3}) + 3(\frac{1}{3}) + 2(-\frac{1}{2}) + 1$	0 ✓
$[SU(3)]^3$	3 and $\bar{3}$ in equal numbers	0 ✓

Why this is remarkable. These are five independent conditions on the hypercharges, and they are satisfied *exactly* — but only when quarks and leptons are counted **together**, with the colour factor of 3 doing essential work in every line. Delete the leptons and the second condition fails; delete the quarks and the third fails; change the number of colours and everything fails.

What it suggests. The hypercharges look arbitrary in the table of Step 1, but they are almost completely fixed by anomaly cancellation. That quarks and leptons need each other for consistency is the strongest structural hint in the Standard Model, and it is the standard entry point to grand unification — where $SU(3) \times SU(2) \times U(1)$ sits inside $SU(5)$ or $SO(10)$, and a single generation fits into one or two irreps with the hypercharges *derived* rather than assigned.

Summary of Lecture 31 — the logic in ten lines

Added explanation (not in the original notes) — Summary

Part I — diagonalising the neutral sector

1. $W^{1,2}$ appear only as $(W^1)^2 + (W^2)^2$, so their mass can be read off at once: $M_W^2 = \frac{1}{4}g_L^2 v^2$. The remaining neutral terms complete a **perfect square**, $\frac{v^2}{8}(g_Y B_\mu - g_L W_\mu^3)^2$ — a rank-one form, so a massless direction is already visible. [p. 7*****]
2. Written as a matrix, $M^2 = \frac{v^2}{4} \begin{pmatrix} g_L^2 & -g_L g_Y \\ -g_L g_Y & g_Y^2 \end{pmatrix}$: non-diagonal, so W^3 and B are *not* mass eigenstates. Diagonalise with an *orthogonal* O (which preserves the canonical kinetic terms), parametrised by the Weinberg angle. [p. 8*****]
3. $\det M^2 = 0$ exactly, so the eigenvalues are $\{0, \frac{v^2}{4}(g_L^2 + g_Y^2)\}$: **one gauge boson stays massless**. This was guaranteed by $QH_0 = 0$ back in Lecture 30. [p. 9*****]
4. The eigenvectors give $O = (g_L^2 + g_Y^2)^{-1/2} \begin{pmatrix} g_L & -g_Y \\ g_Y & g_L \end{pmatrix}$ and hence the mass eigenstates $(Z^\mu, A^\mu)^T = O(W^{3,\mu}, B^\mu)^T$. [p. 10*****]
5. Identifying $\sin \theta_W = g_Y / (g_L^2 + g_Y^2)^{1/2}$, $\cos \theta_W = g_L / (g_L^2 + g_Y^2)^{1/2}$: $Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu$ with $M_Z^2 = \frac{v^2}{4}(g_L^2 + g_Y^2)$, and $A_\mu = \cos \theta_W B_\mu + \sin \theta_W W_\mu^3$ massless. Immediately $M_Z = M_W / \cos \theta_W$, i.e. $\rho = 1$. [p. 11*****]

Part II — charges from the unbroken $U(1)$

6. A_μ is identified with the photon and the unbroken $U(1)$ with electromagnetism. In unitary gauge H points along the vacuum, so h inherits $QH_0 = 0$: **the Higgs is electrically neutral**. [p. 12*****]
7. On the gauge bosons $Y = 0$, so $Q = T^3 + Y = t^3$ — *exactly* the Georgi–Glashow situation. Hence W^3 inert, and $W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2)$ carry charge ± 1 . [p. 13*****]
8. The mode expansion makes this concrete: W_μ^+ destroys a W^+ or creates a W^- , with three polarisations $\lambda = 0, \pm 1$ (the $\lambda = 0$ one being the eaten Goldstone). Z_μ is a combination of two inert fields, hence real and neutral. [p. 14*****]

Part III — spectrum and interactions

9. The ladder: $M_W = \frac{1}{2}g_L v < M_Z = \frac{1}{2}(g_L^2 + g_Y^2)^{1/2} v < M_h = \sqrt{2\lambda} v < v$, with a massless photon at the bottom. Everything is v times a coupling. [p. 15*****]
10. Three categories of bosonic interaction: Higgs self-couplings $(-\lambda v h^3 - \frac{\lambda}{4} h^4)$; Higgs–gauge couplings, which rewrite as $\frac{M_W^2}{v^2}(h^2 + 2hv)W^+W^-$ and $\frac{M_Z^2}{2v^2}(h^2 + 2hv)ZZ$ — *proportional to the vector mass squared*; and gauge self-couplings, whose complete list (W^+W^-A , W^+W^-Z , and four quartics) follows from charge conservation plus the antisymmetry of ε_{ABC} alone. [p. 15*****]–[p. 18*****]

And the opening of the fermionic sector

11. $\Psi = P_L \Psi + P_R \Psi$ with $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$. The Standard Model is *chiral*: left and right components sit in different representations — which forbids every fermion mass term, and forces the Higgs to supply them. [p. 18*****] (*Carried forward in Section 13.*)

Added explanation (not in the original notes) — Where this is going next

The bosonic sector is now complete: masses, mixings, and every interaction among h , $W^\pm$, Z and γ . What remains is the fermionic sector, and it is a bigger job than it looks, because “chiral” is not a detail — it is the organising principle.

Three threads to expect: the **representation assignments** and the charged/neutral current structure they produce (parity violation, and why the photon nonetheless comes out vector-like); the **Yukawa couplings**, which are the only source of fermion mass and which predict $h\bar{f}f \propto m_f/v$; and **anomaly cancellation**, the consistency condition that ties quarks and leptons together and hints at unification.

Beyond that lie the parts of the Standard Model this course has not touched: three generations and the CKM matrix, neutrino masses (which the minimal model does *not* accommodate), and the non-perturbative structure of $SU(3)_c$ — confinement, which no amount of the machinery built in Lectures 24–31 will reach.